WORKED SOLUTION

STEP III 2012, Question 1

Pure • Non-linear Differential Equations

STEP III 2012, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q1

The key idea

Both differential equations are second order and non-linear, so there is no standard recipe. The trick is that the bracket in $\mathrm{d}z/\mathrm{d}x$ can be made to match the left-hand side of each equation by choosing n well. The equation then says $\mathrm{d}z/\mathrm{d}x$ equals something we can integrate at once, which gives a first-order equation we can separate.

Solution

The preliminary result

Write $y' = \mathrm{d}y/\mathrm{d}x$ and $y'' = \mathrm{d}^2y/\mathrm{d}x^2$. By the product rule, using the chain rule on y^n and on $(y')^2$,

$$\begin{aligned}\frac{\mathrm{d}z}{\mathrm{d}x} &= ny^{n-1}y'\,(y')^2 + y^n \cdot 2y'y'' \\ &= y^{n-1}y'(n(y')^2 + 2yy''),\end{aligned}$$

which is the required result.

Part (i)

The left-hand side is $(y')^2 + 2yy''$, which is the bracket with $n = 1$. So take $z = y(y')^2$. Then

$$\frac{\mathrm{d}z}{\mathrm{d}x} = y'((y')^2 + 2yy'') = y'\sqrt{y} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\tfrac{2}{3}y^{3/2}\right).$$

Integrating, $y(y')^2 = \frac{2}{3}y^{3/2} + C$. At $x = 0$ we have $y = 1$ and $y' = 0$, so $0 = \frac{2}{3} + C$ and

$$y\,(y')^2 = \tfrac{2}{3}(y^{3/2} - 1). \qquad (*)$$

Ask yourself $(*)$ only tells us $(y')^2$. Which sign of the square root do we take, and is $y \equiv 1$ (which makes both sides zero) a possibility?

The constant $y \equiv 1$ is *not* a solution of the original equation: it would give $0 = 1$. In fact, putting $x = 0$ into the equation gives $0 + 2y''(0) = 1$, so $y''(0) = \frac{1}{2} > 0$. Hence y has a minimum at $x = 0$ and $y' > 0$ for small $x > 0$. Also the equation is unchanged by $x \mapsto -x$, so the solution is even and we can work with $x > 0$, $y' > 0$, $y > 1$.

Multiply $(*)$ by y, take the positive square root, and separate:

$$\frac{\sqrt{y}\,y'}{\sqrt{y^{3/2} - 1}} = \sqrt{\tfrac{2}{3}}.$$

Substitute $w = y^{3/2} - 1$, so $\mathrm{d}w = \frac{3}{2}y^{1/2}\,\mathrm{d}y$. The left-hand side becomes $\frac{2}{3}w^{-1/2}\,\mathrm{d}w/\mathrm{d}x$, and integrating from $x = 0$ (where $w = 0$),

$$\tfrac{4}{3}\sqrt{w} = \sqrt{\tfrac{2}{3}}\,x \quad \Longrightarrow \quad w = \tfrac{9}{16} \cdot \tfrac{2}{3}\,x^2 = \tfrac{3}{8}x^2.$$

So $y^{3/2} = \frac{3}{8}x^2 + 1$, which by evenness holds for $x < 0$ too. Therefore

$$y = (\tfrac{3}{8}x^2 + 1)^{2/3},$$

as required.

As a direct check, with $u = \frac{3}{8}x^2 + 1$ we get $y' = \frac{2}{3}u^{-1/3} \cdot \frac{3}{4}x = \frac{1}{2}xu^{-1/3}$, so $y(y')^2 = u^{2/3} \cdot \frac{1}{4}x^2u^{-2/3} = \frac{1}{4}x^2$. Meanwhile $\frac{2}{3}(y^{3/2} - 1) = \frac{2}{3} \cdot \frac{3}{8}x^2 = \frac{1}{4}x^2$, so $(*)$ holds.

Part (ii)

Now the left-hand side is $(y')^2 - yy''$. We need $n(y')^2 + 2yy''$ to be a multiple of this, so $n : 2 = 1 : -1$, giving $n = -2$. With $z = y^{-2}(y')^2$ (fine, since y is near 1 and so non-zero),

$$\frac{\mathrm{d}z}{\mathrm{d}x} = y^{-3}y'(-2(y')^2 + 2yy'') = -2y^{-3}y'((y')^2 - yy'') = -2y^{-3}y'\,(-y^2) = \frac{2y'}{y},$$

using the equation in the form $(y')^2 - yy'' = -y^2$. Integrating, $z = 2\ln y + C$, and $x = 0$ gives $0 = 0 + C$. So

$$\left(\frac{y'}{y}\right)^2 = 2\ln y.$$

Let $v = \ln y$, so $v' = y'/y$ and the equation becomes $(v')^2 = 2v$, with $v(0) = 0$.

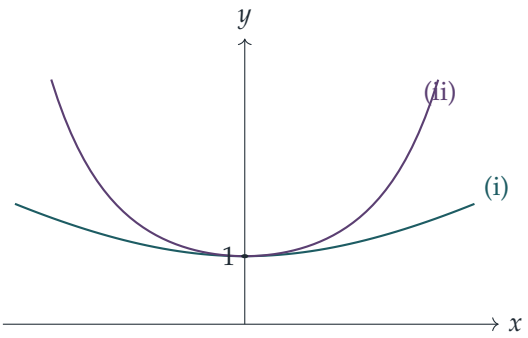
As in part (i), $v \equiv 0$ (that is, $y \equiv 1$) solves this first-order equation but not the original one, which would read $1 = 0$. At $x = 0$ the original equation gives $0 - y''(0) + 1 = 0$, so $y''(0) = 1 > 0$. The equation is again unchanged by $x \mapsto -x$, so we take $x > 0$, where $v' > 0$. Then

$$\frac{v'}{\sqrt{2v}} = 1 \quad \Longrightarrow \quad \sqrt{2v} = x \quad \Longrightarrow \quad v = \tfrac{1}{2}x^2,$$

and by evenness this holds for all x. Hence

$$y = \mathrm{e}^{x^2/2}.$$

To check: $y' = xy$ and $y'' = (1 + x^2)y$, so $(y')^2 - yy'' + y^2 = y^2(x^2 - 1 - x^2 + 1) = 0$. Also $y(0) = 1$ and $y'(0) = 0$.



Checks

- Both solutions are even, with a minimum value of 1 at $x = 0$, as the values of $y''(0)$ predict.
- For part (ii) there is a quicker route: putting $y = \mathrm{e}^v$ straight into the equation gives $y' = v'\mathrm{e}^v$ and $y'' = (v'' + (v')^2)\mathrm{e}^v$, and everything cancels down to $v'' = 1$. So $v = \frac{1}{2}x^2$ immediately.
- Numerically, a Runge–Kutta solution of (i) from $y(0) = 1$, $y'(0) = 0$ gives $y(2) \approx 1.842016$, matching $(\frac{5}{2})^{2/3}$. For (ii) it gives $y(1) \approx 1.648721 = \mathrm{e}^{1/2}$.

Common slips

- Forgetting the constant of integration in z. The initial conditions make it non-zero in part (i): $C = -\frac{2}{3}$.
- Choosing $n = 1$ in part (ii) by analogy. The ratio of the coefficients of $(y')^2$ and yy'' must match, which forces $n = -2$.
- Accepting $y \equiv 1$ from the first-order equation without checking it against the original equation.

Beyond STEP

The method here is an *integrating factor* for a second-order equation. Multiplying $(y')^2 + 2yy''$ by y' turns it into an exact derivative, just as multiplying $\ddot{x} = -\omega^2 x$ by $\dot{x}$ gives conservation of energy. Equations with no explicit x, like these, can always be

reduced to first order by taking $p = y'$ as a function of y. The worry about $y \equiv 1$ is a glimpse of non-uniqueness: $v' = \sqrt{2v}$ fails the Lipschitz condition at $v = 0$. Boyce and DiPrima, *Elementary Differential Equations*, covers both reduction of order and the existence and uniqueness theorem.

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